KATHMANDU UNIVERSITY
End Semester Examination [C]
November, 2018

Marks scored:

Level : B.Sc.
Year : II

(Special compt.)

Course : PHYS 212
Semester : II

Exam Roll No. :

Time: 30 mins.

F. M. : 20

Registration No.:

Date : NOV 15 2018

SECTION "A"
[20 Q × 1 = 20 marks]

Choose and tick the most appropriate answer. The symbols have their usual meanings.

1. Which one of the followings is true for an irreversible process?
[a] $Tds - dU - dw > 0$ [b] $Tds - dU - dw = 0$
[c] $Tds - dU - dw < 0$ [d] $Tds - dU - dw \geq 0$

2. In Carnot's cycle the temperature of the working substance at the end of the cycle is:
[a] equal to zero. [b] more than that at the beginning.
[c] less than that at the beginning. [d] equal to that at the beginning.

3. The change of pressure by one atmosphere changes the melting point of ice by 0.0074^0C and when one gram of ice melts its volume changes by 0.0907 c.c. The latent heat of ice should be equal to
[a] 81.1 cal./gm [b] 8.11 cal./gm
[c] 3.39×10^9cal./gm [d] 33.9×10^9 ergs/gm

4. Which of the following is not a Maxwell thermodynamical relation:
[a] $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$ [b] $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$
[c] $\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$ [d] $\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial T}{\partial V}\right)_P$

5. Which one of the following statements is NOT true?
[a] Joule expansion is a free expansion.
[b] In Joule expansion gas expands in a vacuum.
[c] No external work is done in Joule expansion.
[d] No internal work is done in Joule expansion.

6. Which one of the following statement is wrong?
[a] in a throttling process, the initial and final enthalpies are equal.
[b] Gibb's function remains constant in isothermal processes.
[c] change in entropy during an isobaric process is equal to the heat transferred.
[d] Gibb's function increases in natural processes.

7. The second virial coefficient B
[a] is zero at Boyle temperature
[b] is positive at low temperatures
[c] is higher than first virial coefficient
[d] varies in a different manner for different gases

8. A refrigerator works under a reversible cycle between the temperatures 100 K and 200 K. The coefficient of performance is
[a] 100% [b] 1% [c] 100 [d]1

9. The ratio of Boyle temperature (T_B) to critical temperature (T_c), i.e. $\frac{T_B}{T_c}$, is equal to
[a] $\frac{27}{4}$ [b] $\frac{4}{27}$ [c] $\frac{27}{8}$ [d] $\frac{8}{27}$

10. If coefficient of viscosity, $\eta = 166 \times 10^{-6}$dyne per cm^{2}per unit velocity gradient, $\bar{v} = 4.5 \times 10^4$ cm/s, $\rho = 1.25 \times 10^{-3}$ g/c.c. and $n = 2.7 \times 10^{19}$ molecules per c.c. for nitrogen. The number of collision per second between the molecules is:
[a] 8.8×10^{-6} [b] 3.8×10^{-6} [c] 5.1×10^{9} [d] 5.1×10^{-9}

11. The average energy of the molecules of monatomic gas at temperature T is
[a] $\frac{1}{2}$kT [b] kT [c] $\frac{3}{2}$kT [d] $\frac{5}{2}$kT

12. Five particles are distributed in two phase cells. The number of macrostates is:
[a] 3 [b] 6 [c] 12 [d] 32

13. According to Boltzmann canonical distribution law:
[a] low energy cells contain more particles.
[b] high energy cells contain more particles.
[c] number of molecules having zero energy is greater than any other energy.
[d] zero energy molecules are zero.

14. According to Wein's law:
[a] $\lambda_m T =$ constant [b] $\frac{\lambda_m}{T} =$ constant
[c] $\frac{T}{\lambda_m} =$ constant [d] $\lambda_m^2 T =$ constant

15. The minimum vibrational energy of an oscillator is:
[a] 0 [b] $h\nu$ [c] $\frac{1}{2}h\nu$ [d] $2h\nu$

Fill in the blanks.

16. Entropy remains constant in.....process.

17. The number of degree of freedom of each molecule of a monatomic gas is .....

18. The particles obeying Maxwell-Boltzmann statistics are .....

19. Two bodies are at temperatures 27°C and 927°C. The heat energy radiated by them will be in the ratio.....

20. The internal energy of an ideal gas depends only upon.....

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SECTION "B"

[5Q.×4=20 marks]

1. Starting from Maxwell's Thermodynamic relations obtain Clausius-Clapeyron equations and also show that the ratio of adiabatic and isothermal elasticities is equal to $\frac{C_P}{C_V}$.

2. The spectral distribution of black-body radiation at temperature T K is given by Planck's law

$$U_\nu d\nu = \left(\frac{8\pi\nu^2 d\nu}{c^3}\right)\left(\frac{h\nu}{e^{\frac{h\nu}{kT}} - 1}\right)$$

Obtain the Stefan's constant (σ) and Wien's constant (b)

3. What do you mean by thermodynamical probability? Deduce Boltzmann's entropy-probability relation $S = K \log_e \Omega$.

OR

Prove that according to law of equipartition of energy, the energy associated with each degree of freedom is $\frac{1}{2}kT$.

4. What do you understand by transport properties? Derive an expression for coefficient of viscosity of a gas in terms of mean free path of its molecules.

5. Calculate the Fermi energy in electron volts for sodium assuming that it has one free electron per atom. Given density of sodium = 0.97gm/cm^3, atomic weight = 23.

OR

The collision diameters of hydrogen, nitrogen and chlorine gases are 2.5 Å, 3.5 Å and 4.0Å respectively. Compare their thermal conductivities.

SECTION "C"

[5Q.×7=35 marks]

6. What is Joule-Thomson effect? How this expansion differs from adiabatic and Joule expansion. In case of Joule-Thomson effect, verify the relation:

$$dT = \frac{1}{C_p}\left[\frac{2a}{RT} - b\right]dP$$

Where dT gives the change in temperature of a Vander Waal gas due to Joule Thomson expansion.

7. Derive an expression for Boyle temperature, temperature of inversion and critical temperature and hence show that $T_i = 2T_B = 6.75\, T_C$.

8. At what temperature will the average speed of hydrogen molecules be the same as that of nitrogen molecules kept at 35^{0}C. Molecular weights of nitrogen and hydrogen are 28 and 2 atomic mass units respectively. Also calculate the most probable speed, root mean square speed and average speed of nitrogen molecules at this temperature. If pressure corresponding to this temperature is 10^5N/m^2, calculate the density of nitrogen.
[Given: Boltzmann constant = 1.38×10^{-23} joules per degree.]

OR

The coefficient of viscosity of Helium at 273K and one atmosphere is η_1, that of Argon is η_2. The atomic masses of these monoatomic gases are μ_1 and μ_2 respectively. Calculate in terms of the above constants:

a. The ratio $\frac{S_2}{S_1}$, where S_2 is the total scattering cross-section of the Ar-Ar atom while S_1 is the total scattering cross-section of the He-He atom.

b. The ratio $\frac{K_2}{K_1}$, where K_2 is the thermal conductivity of Argon gas, while K_1 is the thermal conductivity of the Helium gas at T=273K.

9. Compare the basic postulates of the Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac statistics. Deduce the law of distribution of energy of particles according to Fermi-Dirac statistics.

10. Write down the assumptions of Einstein's theory of specific heat capacity of solid. Derive an expression for lattice heat capacity for this model. Also discuss the limitation of this theory.